Author Response to Reviewers

Journal: Expert Systems (Wiley Blackwell)

Manuscript Reference: 2961684 (Re-Revision / Minor Revisions)

Manuscript Title: Legalyze: An Open-Source Structure-Aware Agentic RAG Platform for High-Precision Legal Intelligence

Authors: Ali Hamza, Maaz Bin Rizwan, Maemoona Kayani, Shafa-at Ali Sheikh

---

Dear Editors and Reviewers,

We sincerely thank the editorial team, Reviewer 1, and Reviewer 2 for their time, thorough evaluation, and constructive feedback on our manuscript. The reviewers' insights have been invaluable in enhancing the statistical rigor, methodological transparency, scope clarity, and domain contextualization of Legalyze. We have fully addressed every comment raised in detail below, providing exact point-by-point responses and pinpointing the precise page, section, table, and paragraph locations of all corresponding revisions in the manuscript.

REVIEWER 1

The authors successfully addressed my previous concerns.

Author's Response: We sincerely thank Reviewer 1 for their positive assessment, supportive feedback, and confirmation that all previous concerns have been satisfactorily resolved.

REVIEWER 2

We are deeply grateful to Reviewer 2 for the positive and encouraging assessment of Legalyze's architecture, open-source contribution, and relevance to jurisdiction-specific legal AI. We address all methodological, statistical, and textual comments below.

Comment 1: Integration of tables, figures, and cross-references

1. Please complete the integration of tables, figures, and cross-references.
The revised manuscript still contains several unresolved placeholders, including "??," "Table 1 goes here," "Table 3 goes here," and similar statements for Tables 2, 4, and 5. These appear in the literature review, evaluation framework, Results section, and retrieval ablation section. All tables should be inserted into the main manuscript, numbered consistently, and cited at the appropriate locations. The same applies to the references to the qualitative comparison and domain-level results.
This is primarily a production issue, but it currently interrupts the flow of the paper and makes some claims difficult to trace.

Author's Response: We sincerely thank the reviewer for this vital feedback. In the revised manuscript, all five tables (Tables 1–5) and all figures (Figures 1–8) have been fully integrated into the manuscript body at their appropriate positions with complete cross-referencing, resolving all placeholders and broken references.

In the previous manuscript:

Unresolved table placeholders ('Table 1 goes here', etc.) and broken cross-references ('??') appeared in the text.

In the revised manuscript (Table 1, p. 25; Table 2, p. 14; Table 3, p. 17; Table 4, p. 26; Table 5, p. 20; Figures 1–8, pp. 6, 9, 11, 12, 16):

All tables and figures have been fully integrated into the manuscript body, with all cross-references resolving correctly.

Comment 2: Paired and repeated-measures statistical analysis (Cochran's Q, McNemar, GEE)

2. The statistical analysis should account for the paired and repeated-measures design.
The same 67 queries were evaluated under all three system configurations, and the same three reviewers rated the corresponding outputs. The observations are therefore paired rather than independent. Pearson's chi-squared test of independence does not fully reflect this structure.
For the consensus labels, Cochran's Q test followed by paired McNemar tests would be more appropriate. For the pooled reviewer judgments, a mixed-effects logistic model or a generalized estimating equation could account for clustering by query and reviewer. This reanalysis should be possible using the existing annotations and would not require additional data collection.
Please also report exact p-values for all pairwise comparisons rather than only $p<0.05$, and briefly explain how the overall Fleiss' $\kappa=0.743$ was calculated, particularly because it is higher than each of the three configuration-specific values.

Author's Response: We express our deepest gratitude to the reviewer for this invaluable and constructive statistical guidance. We completely agree that because the identical 67 legal queries were evaluated across all three systems, the data possesses a paired/repeated-measures structure. We have replaced the independent Pearson Chi-squared test with Cochran's Q test (omnibus), pairwise exact binomial McNemar tests, and a Generalized Estimating Equations (GEE) logistic regression model on the pooled reviewer judgments to account for clustering by query. We outline these statistical methods in Section 3.4 (Experimental Setup), report exact p-values and odds ratios in Section 4.1 (End-to-End Hallucination Evaluation), and provide a formal statistical note explaining why the overall Fleiss' Kappa is higher than per-configuration values due to the prevalence effect.

In the previous manuscript:

"Pearson's Chi-squared (χ^2) tests of independence were performed for each system configuration (Zero-Shot vs. Legalyze: $\chi^2 = 41.20$, $p < 0.0001$). No paired or clustered model was reported."

In the revised manuscript (Abstract, p. 1–2; Introduction, p. 3; Section 3.4 Experimental Setup, p. 15; Section 4.1 End-to-End Hallucination Evaluation, p. 18):

Abstract (p. 1–2): 'The improvement over both baselines is statistically significant across all pairwise comparisons (Cochran's Q = 37.57, p = 6.93 x 10^-9; McNemar exact: Zero-Shot vs. Legalyze p = 5.40 x 10^-9).'

Introduction (p. 3): 'Combined with a dual-stream retrieval pipeline and a multi-agent reasoning architecture, Legalyze is designed to ground generated claims in verifiable statutory evidence. The evaluation methodology follows a reviewer-blinded protocol over 67 legal queries evaluated across three system configurations, with hallucination traceability independently assessed by three legal reviewers; statistical significance across all paired comparisons is established using omnibus Cochran's Q tests, pairwise exact McNemar tests, and Generalized Estimating Equations (GEE) modeling to account for query clustering.'

Section 3.4 Experimental Setup (p. 15): 'To evaluate the empirical findings with statistical rigor, four complementary inferential methods are employed: 1) Inter-Rater Reliability (Fleiss' Kappa, kappa) is

calculated across the three independent legal reviewers within each system configuration to quantify multi-rater agreement on the binary traceability criteria. 2) Omnibus Paired Hypothesis Testing (Cochran's Q Test) is conducted on the N = 67 paired, repeated-measures consensus judgments across the three system configurations (Zero-Shot, Standard RAG, and Legalyze) to establish whether overall differences in hallucination rates are statistically significant without violating observational independence. 3) Pairwise Contrasts (Exact Binomial McNemar Tests) are applied to each paired 2 x 2 system contingency table, calculating exact two-tailed p-values to evaluate the statistical significance of pairwise reductions in legal hallucinations. 4) Clustered Binary Regression (Generalized Estimating Equations - GEE) is fitted on the full pooled dataset (N = 603 total judgments) using a binomial logit link, exchangeable working correlation structure, and robust Huber-White sandwich standard errors to estimate adjusted odds ratios (OR) and confidence intervals while formally accounting for within-query repeated-measures clustering across reviewers.'

Section 4.1 End-to-End Hallucination Evaluation (p. 18): 'Cochran's Q Test (consensus-adjudicated dataset, N = 67 queries): Q = 37.57, df = 2, p = 6.93 x 10^-9 (extremely significant across all three configurations simultaneously).
Pairwise McNemar Tests (exact binomial; all pairwise comparisons significant):
• Zero-Shot vs. Standard RAG: p = 7.53 x 10^-4
• Zero-Shot vs. Legalyze: p = 5.40 x 10^-9
• Standard RAG vs. Legalyze: p = 4.43 x 10^-3
GEE Analysis (pooled, N = 603 judgments, 67 query clusters, Binomial family, exchangeable within-query correlation, robust sandwich SE; reference category = Legalyze):
• Standard RAG vs. Legalyze: OR = 5.13 [95% CI: 1.76, 14.92], p = 0.003
• Zero-Shot vs. Legalyze: OR = 20.88 [95% CI: 7.29, 59.81], p = 1.52 x 10^-8
The GEE results confirm that, after accounting for repeated measurements across the same 67 queries, Standard RAG has 5.1x higher odds of hallucination than Legalyze, and Zero-Shot has 20.9x higher odds; both are highly statistically significant.'

Note on Fleiss' Kappa (Section 4.1, p. 18): 'Note on Fleiss' Kappa. Per-configuration Fleiss' kappa values (Zero-Shot: kappa = 0.659; Standard RAG: kappa = 0.718; Legalyze: kappa = 0.615) are lower than the overall pooled kappa = 0.743. This is a well-documented statistical phenomenon known as the prevalence effect (Feinstein & Cicchetti, 1990): when one category dominates within a single configuration (e.g., 94.5% "Y" for Legalyze), expected agreement by chance (Pe) is very high, which deflates the per-configuration kappa. Pooling all three configurations produces a more balanced Y/N distribution, which reduces Pe and yields a higher overall kappa. This is not an inconsistency; it is an inherent mathematical property of kappa under skewed distributions and does not affect the reliability interpretation.'

Comment 3: Precision in evaluation terminology and unit of analysis

3. The terms "double-blind," "answer," "statement," and "judgment" need to be used more precisely.
The evaluation is described as "strict double-blind." However, the researchers necessarily knew how the outputs were produced, and reviewers could potentially recognize the zero-shot condition because it had no retrieved passage. "Reviewer-blinded evaluation" or "blinded expert assessment" would be a more accurate description. The authors should also state whether a neutral placeholder was displayed when no retrieval was used.
There is also some ambiguity in the unit of analysis. The manuscript refers to 201 generated "statements," although these appear to be 67 responses from each of three systems. These responses then produce 201 reviewer judgments per system and 603 judgments overall. Please state clearly whether each response received one binary label or whether multi-sentence responses were first divided into individual legal propositions. The terminology and sample sizes should remain consistent throughout the Abstract, Methods, Results, tables, and figure captions.

Author's Response: We sincerely thank the reviewer for this insightful recommendation on precision and clarity in evaluation terminology. We have replaced 'strict double-blind' with 'reviewer-blinded evaluation protocol' throughout the manuscript and added an explanatory footnote clarifying that reviewers were blinded to system identities while the research team necessarily knew them. We have also explicitly described the neutral placeholder text displayed for Zero-Shot evaluations and unified the unit of analysis to 'responses' throughout the entire manuscript (Abstract, Methods, Results, tables, and captions). The evaluation comprises 201 system responses (from 67 queries x 3 systems), each assigned a single binary traceability label by 3 reviewers, yielding 201 individual judgments per system, 603 judgments overall, and a consensus-adjudicated dataset of 67 unique response labels per system.

In the previous manuscript:

"Assessed all 201 statements under a strict double-blind protocol: reviewers saw only the query, the retrieved source passage, and the generated answer."

In the revised manuscript (Section 3.4 Experimental Setup, p. 15; Section 4 Results, p. 16):

Section 3.4 Experimental Setup (p. 15): 'Three independent qualified legal reviewers, active Advocates of the High Court of Pakistan each with between five and eight years of litigation experience in Pakistan's superior courts, assessed all 201 responses under a reviewer-blinded evaluation protocol [Footnote 1: The term "blinded" is used in the sense that reviewers were blinded to the identity of the source system; the research team necessarily knew which configuration generated each response. This follows standard practice for human-subject evaluation studies.]: reviewers saw only the query, the retrieved source passage (if any), and the generated response. For Zero-Shot responses, reviewers were shown the placeholder text "(No retrieval passage: this response was generated without source context.)". The system identity mapping (System A/B/C) was decoded only after all reviews were complete. Reviewers independently annotated each response using the binary traceability criterion: a response is marked as hallucinated (N) if the reviewer cannot locate supporting text in either the retrieved passages or the original statutory corpus; otherwise it is marked as traceable (Y). In a subsequent joint session, any disagreements were resolved by majority discussion to produce a single

consensus label per response, yielding two datasets: a pooled dataset of 201 individual judgments per configuration and a consensus-adjudicated dataset of 67 unique response labels per configuration.'

Section 4 Results (p. 16): 'A total of 201 generated responses (67 legal queries x 3 system configurations) were assessed by three independent qualified legal reviewers under a reviewer-blinded evaluation protocol, as described in Section 3.4. Each response received a single binary traceability label from each reviewer.'

Comment 4: Reconciliation of evaluation corpus descriptions

4. Please reconcile the descriptions of the evaluation corpus.
The main manuscript describes the corpus as consisting of the Constitution of Pakistan, the Pakistan Penal Code, and 250 superior-court judgments. By contrast, the supplementary corpus sample also lists the Code of Criminal Procedure and the Qanun-e-Shahadat Order. Since the evaluation contains a procedural-law category, the role of the procedural statutes should be made explicit.
Please provide one definitive corpus description, including the documents used for indexing, the documents used for query construction, and the token contribution of each source. In addition, the phrase "drawn uniformly from the three corpora" should be revised or explained, because the domain totals in Table 3 correspond to 24 constitutional, 23 criminal, and 20 procedural queries rather than equal-sized groups.

Author's Response: We are very grateful to the reviewer for this helpful comment and for prompting a unified, complete reconciliation of our evaluation corpus. We have updated the text in both Section 3.3 (Analysis) and Section 3.4 (Experimental Setup) to provide a complete, definitive enumeration of all five Pakistani legal sources (Constitution, PPC, CrPC, QSO, and 250 judgments) and their exact token contributions. We have also updated Figure 7 to illustrate the complete 5-corpus breakdown and replaced the phrase 'drawn uniformly' with the exact query count and percentage breakdown across the three legal domains.

In the previous manuscript:

"Three Pakistani legal corpora (Constitution, PPC, and 250 superior court rulings) were used. 67 legal queries drawn uniformly from the three corpora."

In the revised manuscript (Section 3.3 Analysis, p. 13; Section 3.4 Experimental Setup, p. 14; Figure 7 Caption, p. 16):

Section 3.3 Analysis (p. 13): 'Legalyze is analyzed using mathematical models and empirical data. Five Pakistani legal sources were used to build the evaluation corpus (Figure 7): the Constitution of Pakistan (1973, 280 articles and six schedules), the Pakistan Penal Code (PPC, 1860, 511 sections), the Code of Criminal Procedure (CrPC, 1898), the Qanun-e-Shahadat Order (QSO, 1984, Pakistan's primary evidence statute), and 250 superior court rulings spanning 2018 to 2024. These five sources jointly contain a total of approximately 1.2 million tokens of legal text, with statutory legislation contributing roughly 54% (~648,000 tokens) and the case-law collection comprising the remaining 46% (~552,000 tokens). To ensure rigorous and comprehensive empirical assessment, the evaluation directly maps these sources into three legal domains: 1) Constitutional Law (24 queries, 72 reviewer judgments), evaluated against the Constitution of Pakistan (1973); 2) Criminal Law (23 queries, 69 reviewer judgments), evaluated against the Pakistan Penal Code (PPC, 1860); and 3) Procedural Law (20 queries, 60 reviewer judgments), evaluated jointly against the Code of Criminal Procedure (CrPC, 1898) for procedural rules and the Qanun-e-Shahadat Order (QSO, 1984) for evidentiary admissibility and burden-of-proof standards. Across all three domains, the 250 superior court judgments provide the contextual judicial interpretation layer and precedent hierarchy...'

Section 3.4 Experimental Setup (p. 14): 'The hallucination study was conducted on 67 legal queries drawn from three legal domains: Constitutional Law (24 queries, 35.8%, targeting the Constitution of Pakistan), Criminal Law/PPC (23 queries, 34.3%, targeting the Pakistan Penal Code), and Procedural

Law (20 queries, 29.9%, jointly drawing upon the Code of Criminal Procedure [CrPC] for procedural rules and the Qanun-e-Shahadat Order [QSO] for evidentiary admissibility standards, with the 250 superior court judgments providing contextual judicial interpretation across all three domains). Each query was executed on all three systems, yielding 201 generated responses (67 x 3 = 201). Each response received a single binary traceability label from each reviewer, yielding 201 individual judgments per configuration and 603 judgments in total. The same 67 queries were used across all three configurations to ensure a paired and directly comparable design.'

Figure 7 Caption (p. 16): 'Test Corpus Distribution by Document Type. The evaluation corpus consists of approximately 1.2 million tokens drawn from five Pakistani legal sources: Constitution of Pakistan, PPC, CrPC, Qanun-e-Shahadat Order, and 250 superior court judgments. Constitutional statutes account for roughly 54% and the case-law collection for the remaining 46%.'

Comment 5: Distinguishing offline retrieval ablation from production retriever

5. The offline retrieval ablation should be distinguished from the deployed semantic retriever.
The operational system is described as using sentence embeddings and Pinecone vector search. In the offline ablation, however, the dense stream is represented by a character 3-gram vector-space model. This is a reasonable deterministic proxy, but it is not identical to the deployed dense retrieval component.
The manuscript should therefore make clear that Table 5 evaluates an offline surrogate pipeline. The configuration currently called "Legalyze Full" could be renamed "Offline SAHI Hybrid with RRF," or the difference should be stated prominently in the table caption and discussion. Similarly, the Abstract should not imply that the reported Recall@5 value directly measures the production Pinecone pipeline unless the same embedding model and retrieval settings were used.

Author's Response: We sincerely thank the reviewer for this important methodological distinction between our offline ablation study and the production retriever. The local 7-configuration retrieval ablation utilizes a character 3-gram VSM cosine similarity model as a local deterministic proxy. We have renamed the baseline in Table 5 to 'SAHI Hybrid + RRF†', prominently stated the surrogate nature in the table caption and footnote, and updated all prose references in Section 4.3 to refer specifically to 'the offline SAHI Hybrid with RRF configuration', ensuring total consistency between the table and text.

In the previous manuscript:

"Row 7 in Table 5 was labeled 'Legalyze Full (Proposed)' and prose read 'Combining the SAHI streams under Legalyze Full achieves...'."

In the revised manuscript (Abstract, p. 1–2; Section 4.3 Retrieval Ablation Study, p. 19; Table 5, p. 20):

Abstract (p. 1–2): 'A seven-configuration offline retrieval ablation further demonstrates that an offline surrogate of the Legalyze architecture achieves Recall@5 of 65.0%, compared to 45.0% for flat lexical retrieval, confirming the retrieval gains attributable to structure-aware indexing...'

Section 4.3 Retrieval Ablation Study (p. 19): 'In this offline study, the dense retrieval stream is approximated by a character 3-gram vector space model (VSM) rather than the production Pinecone sentence-embedding model, which constitutes a conservative local proxy. Full online evaluation with the production embedding model is planned as future work... Combining the SAHI streams under the offline SAHI Hybrid with RRF configuration achieves the highest Recall@5 of 65.0%.'

Table 5 Caption & Footnote (p. 20): 'Retrieval Ablation Results across seven configurations (20 ground-truth queries). † All configurations use character 3-gram cosine similarity as the dense retrieval proxy; the deployed production system uses Pinecone sentence embeddings. These results therefore characterise the offline surrogate pipeline.'

Comment 6: Cross-jurisdictional citations and scope of claims

6. Please address minor citation, scope, and formatting inconsistencies.
(a) Citations to Fang et al. should be distinguished: The bibliography contains multiple entries for Fang et al. (2024/2025). The in-text citations in the Introduction and Discussion should be disambiguated so that each reference points to the intended paper.
(b) The scope of regional generalizability claims should match the evaluated corpus: In several places, the discussion suggests broader applicability to South Asian or common-law jurisdictions. While this is a plausible extension, the empirical evaluation is limited to Pakistani statutes. The text should consistently reflect this scope, particularly in the Introduction and Discussion.
(c) Minor typographical and reference-formatting issues: Please check the bibliography for duplicate entries, missing conference/journal details, and inconsistent author name formats.

Author's Response: We are very grateful to the reviewer for these helpful observations. We have carefully disambiguated all six citations to Fang et al. across the Introduction, Literature Review, and Discussion. Furthermore, we revised our discussion on regional generalizability in Section 5 (Discussion), explicitly clarifying that Legalyze is currently designed and evaluated specifically for the Pakistani legal system (Constitution, PPC, CrPC, QSO, and 250 judgments), while framing multi-jurisdictional expansion as part of our future roadmap. Finally, we conducted a thorough audit of the bibliography to eliminate duplicate entries and ensure full formatting consistency.

In the previous manuscript:

"Six papers by Fang et al. (2024a-f) had overlapping citation keys, and regional generalizability suggested immediate cross-jurisdictional applicability."

In the revised manuscript (Section 5 Discussion, p. 21–22):

Section 5 Discussion (p. 21–22): 'Furthermore, the present release (v1.0) of Legalyze exclusively supports the Pakistani legal jurisdiction; the corpus, statutory hierarchy, and jurisdictional filtering mechanisms have been designed and validated specifically across all five primary sources: the Constitution of Pakistan (1973), the Pakistan Penal Code (1860), the Code of Criminal Procedure (1898), the Qanun-e-Shahadat Order (1984), and the collection of 250 superior court judgments. Extending coverage to neighbouring South Asian legal systems, including Indian, Bangladeshi, and Nepalese statutory frameworks, is planned as a structured roadmap item for subsequent versions, as each jurisdiction requires its own corpus ingestion pipeline, heading-detection rules, and precedent adjudication hierarchy...'

Comment 7: Moderation of promotional claims and scope boundaries

7. Several claims should be moderated in view of the current evaluation scope.
The transparent reporting of a 35.7% exact SAHI parsing accuracy is appreciated. At the same time, statements such as "professional-grade," "suitable for professional use," "ensures that every claim is grounded," and "provides a new standard" are stronger than the current evidence supports. The evaluation mainly measures source traceability, uses a relatively small query set, and does not yet report the legal-correctness and citation metrics introduced in Section 3.3.
I suggest presenting Legalyze as a promising legal research and decision-support prototype that requires professional verification, rather than as a system already validated for autonomous professional use. The discussion should also avoid attributing the domain-level differences solely to hierarchical indexing unless a corresponding domain-specific ablation supports that interpretation.

Author's Response: We are deeply grateful to the reviewer for this constructive advice on moderating promotional claims and emphasizing rigorous academic precision. We have thoroughly revised all instances of assertive marketing phrasing throughout the paper, replacing them with measured, research-oriented terminology ('5.5% pooled hallucination rate', 'promising for use as a legal research and decision-support aid under human expert verification', 'is designed to ground generated claims in verifiable statutory evidence'). We have also added explicit scope boundaries distinguishing grounding traceability from substantive legal correctness.

In the previous manuscript:

"Claims of 'research-grade 5.5% HR', 'ensures that all outputs are highly traceable', 'allow legal practitioners to use AI without affecting quality in high-stakes settings', 'providing a strong empirical foundation for its use in professional legal practice', and 'ground every output'."

In the revised manuscript (Section 1 Introduction, p. 3; Section 4 Results, p. 17; Section 5 Discussion, p. 21; Section 6 Conclusion, p. 23):

Section 1 Introduction (p. 3): 'Combined with a dual-stream retrieval pipeline and a multi-agent reasoning architecture, Legalyze is designed to ground generated claims in verifiable statutory evidence.'

Section 4 Results (p. 17): '...providing encouraging empirical evidence for its potential as a legal research and decision-support aid under human expert verification.'

Section 5 Discussion (p. 21): '...the overall pooled HR (4.5% consensus-adjudicated) demonstrates that, with structure-aware retrieval, the system can search through 1.2 million tokens of the Pakistani legal corpus with a level of accuracy that is promising for use as a legal research and decision-support aid under human expert verification.'

Section 6 Conclusion (p. 23): 'The platform relies on a retrieval-augmented architecture that is intended to preserve structural knowledge and is designed to ground generated outputs in verifiable legal sources. These results suggest potential for Legalyze as a research and decision-support aid, subject to human legal verification in high-stakes settings.'

CLOSING REMARKS & SIGN-OFF

We once again express our sincere gratitude to the Editor, Reviewer 1, and Reviewer 2 for their constructive reviews. We believe that these rigorous statistical, methodological, and textual enhancements make the manuscript substantially stronger and ready for publication in Expert Systems.

Corresponding Author:

Ali Hamza
Department of Software Engineering, National University of Sciences and Technology (NUST), Islamabad, Pakistan
Email: ahamza.msse25mcs@student.nust.edu.pk

On behalf of all co-authors: Maaz Bin Rizwan, Maemoona Kayani, Shafa-at Ali Sheikh